

SURAJ KUMAR GUPTA

Full-Stack Engineer | AI/ML Engineer

Bhubaneswar, India | +91 7070031833 | surajgupta7070031833@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Full-Stack Software Engineer and AI/ML Engineer with hands-on experience building and shipping production web applications (MERN, Next.js, TypeScript) and Generative AI systems (LLM integration, RAG pipelines, agentic workflows, vector search, MLOps). Skilled in REST API design, relational and NoSQL databases, authentication, and cloud deployment on AWS, Vercel, and Render. Strong Data Structures & Algorithms foundation with 500+ problems solved.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, Shadcn UI, Framer Motion, Responsive Design

Backend: Node.js, Express.js, Bun, FastAPI, REST APIs, JWT Auth & Authorization, Clerk, Webhooks, Background Jobs (Inngest)

Databases: PostgreSQL (Neon), MongoDB, Prisma ORM, Redis, Supabase, Vector Databases (FAISS, ChromaDB)

Generative AI: LLMs, RAG, Hybrid-RAG, Agentic Workflows, LangChain, LangGraph, Prompt Engineering, Embeddings, OpenAI API, Gemini API

AI/ML: Machine Learning, NLP, Transformers, Scikit-Learn, TensorFlow, MLOps, Model Deployment

Cloud & DevOps: CI/CD, Docker, GitHub Actions, AWS (EC2, ECR), Vercel, Render, Appwrite, Git, Linux

Concepts: REST API Design, Role-Based Access Control (RBAC), System Design, Database Design, Data Structures & Algorithms, Agile

EXPERIENCE

Independent Full-Stack & AI Engineer | *Self-Directed / Independent Projects*

2024 – Present

- Designed, built, and deployed 8+ production web and AI applications on Vercel, Render, and AWS using Next.js, React, TypeScript, Node.js, Python, FastAPI, PostgreSQL, and MongoDB.
- Engineered REST APIs with JWT authentication, role-based authorization, Stripe payments, background jobs, and third-party integrations for multi-vendor e-commerce and hyperlocal delivery platforms.
- Developed Generative AI applications with LLMs, RAG, Hybrid-RAG, vector search, embeddings, LangChain, and LangGraph.
- Automated CI/CD and MLOps workflows with Docker, GitHub Actions, AWS ECR, and EC2, plus Streamlit-based ML dashboards.

PROJECTS

GoCart – Multi-Vendor E-Commerce Platform

[Live](#) | [GitHub](#)

Next.js 15, React 19, Tailwind CSS, Clerk, Prisma, Neon PostgreSQL, Stripe, ImageKit, Inngest

- Built a multi-vendor marketplace on Next.js App Router with buyer, seller, and admin roles, Clerk authentication, and an admin-reviewed seller onboarding workflow.
- Shipped Stripe/COD checkout, seller order and stock dashboards, and an OpenAI-powered listing assistant that auto-generates product titles and descriptions from images.

FreshCart – Full-Stack Grocery Delivery Application

[Live](#) | [GitHub](#)

React 19, TypeScript, Bun, Express, Prisma, PostgreSQL (Neon), Stripe, Cloudinary, Inngest

- Built a full-stack hyperlocal grocery delivery platform with dedicated customer, store-admin, and delivery-partner portals, using JWT authentication and role-based access control.
- Implemented Stripe checkout with webhook-verified payments, live Leaflet-based order tracking, OTP-verified delivery handoff, Cloudinary image uploads, and Inngest background email jobs.

Hybrid-RAG – Advanced AI Retrieval System

[GitHub](#)

Python, Gemini, ChromaDB, NetworkX, Knowledge Graphs

- Built a local-first hybrid RAG pipeline combining dense vector search (ChromaDB) with knowledge-graph traversal (NetworkX) to answer semantic and multi-hop queries.
- Automated knowledge-graph construction via LLM-based triple extraction and built an LLM query router that dynamically selects vector, graph, or hybrid retrieval.

Invoice Intelligence Dashboard – ML-Powered Finance Platform

[Live](#) | [GitHub](#)

Python, Streamlit, Scikit-Learn, Plotly, Pandas

- Built an ML-powered Streamlit platform with a Linear Regression model for freight-cost prediction and a Random Forest classifier that flags high-risk invoices for manual review.
- Delivered interactive Plotly visualizations for cost forecasting, reducing financial leakage and speeding up invoice approval decisions.

ML Pipeline with CI/CD – End-to-End MLOps

[GitHub](#)

Python, Scikit-Learn, Docker, GitHub Actions, AWS EC2, AWS ECR

- Built an end-to-end ML pipeline for student performance prediction, covering data processing through model serving.
- Automated containerization and cloud delivery using GitHub Actions, Docker, and AWS ECR/EC2, achieving zero-downtime deployments.

EDUCATION

B.Tech in Computer Science & Engineering – KIIT University, Bhubaneswar

Expected May 2027

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

KEY ACHIEVEMENTS

- Solved 500+ Data Structures & Algorithms problems on LeetCode and Codeforces, covering graphs, dynamic programming, and greedy algorithms.
- Independently built and deployed 8+ production applications across Vercel, Render, and AWS, including a GPT-2 model built from scratch with a custom tokenizer, multi-head attention, and autoregressive generation.